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Indian Institute of Technology Hyderabad

Programming for AI

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Rejection Sampling (Accept-Reject) Technique

- Many a times, computing F^{-1} is tedious
- For instance, consider the PDF

$$f(x) = 6x(1-x), \qquad x \in [0,1]$$

- The CDF corresponding to the above PDF is given by

$$F(x) = \begin{cases} 0, & x < 0, \\ 3x^2 - 2x^3, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

- Computation of F^{-1} is non-trivial; need to use numerical techniques
- **Alternative solution:** Rejection sampling

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Rejection Sampling (Accept-Reject) Technique

Objective

Given a PDF f with CDF F for which computing F^{-1} is tedious, generate a sample $X \sim f$.

- Let U, Z be a **pair of continuous random variables** satisfying the following properties:
  1. $U \sim \mathrm{Unif}(0,1)$, Z has PDF f_Z, **easy to sample $Z \sim f_Z$**
  2. U is independent of Z
  3. There exists a real number $a \geq 1$ such that

$$f(x) \leq a\, f_Z(x) \qquad \forall\, x \in \mathbb{R}.$$

- Generate (U, Z) satisfying 1-3 above
  - (i) If $a\, U f_Z(U) \leq f(U)$, **accept** (U, Z), and set $X = Z$
  - (ii) If not, **reject** current (U, Z) and generate new (U, Z)
  - (iii) Repeat till (i) holds

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Theorem (Rejection Sampling)

Let $E = \{a\, U\, f_Z(Z) \leq f(Z)\}$. Then,

$$F_X(x) \;=\; \mathbb{P}(\{X \leq x\}) \;=\; \mathbb{P}(\{Z \leq x\} \mid E).$$

Example

- Use rejection sampling to generate a sample $X \sim f$, where

$$f(x) = 6x(1-x), \qquad x \in [0, 1].$$

Example

- For fixed constants $\lambda, t > 0$, the $\mathrm{Gamma}(\lambda, t)$ PDF is given by

$$f(x) \;=\; e^{-\lambda x}\,\frac{\lambda^t x^{t-1}}{\Gamma(t)}, \qquad x > 0.$$

1. When $t \in \mathbb{N}$, suggest a technique to generate a sample $X \sim f$ via ITT.
2. When $t \notin \mathbb{N}$, design an algorithm to sample $X \sim f$ via rejection sampling.
   Hint: Take $Z \sim \mathrm{Exponential}(1/t)$.

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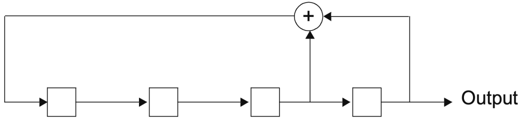
Pseudo-Random Number Generators (PRNGs)

Binary PRNGs

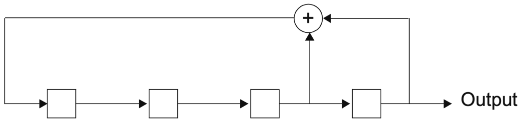


$S_0S_1S_2S_3$	Output
1111	1
0111	1
0011	1
0001	1
1000	0
0100	0
0010	0

$S_0S_1S_2S_3$	Output
1001	1
1100	0
0110	0
1011	1
0101	1
1010	0
1101	1
1110	0

Output (one period): 1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0

Properties of the Binary PRNG



Output (one period): $1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0$

- Number of zeros in one period $\approx$ number of ones in one period
  (**desirable** of uniform binary random number generator)
- Period = 15
  (**not desirable** of uniform binary random number generator)

Possible Workaround for Periodicity in Output

Increase the number of stages N.

N-Stage Binary PRNG

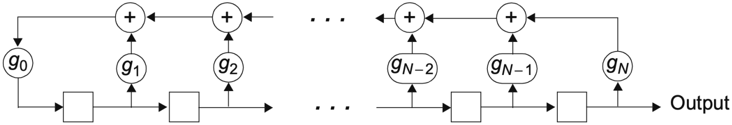


Figure: N-Stage, binary linear feedback shift register.

- $g_0 = g_N = 1$
- Adjust the tap gains $\{g_1, \ldots, g_{N-1}\}$ to achieve highest possible period $(= 2^N - 1)$
- E.g., for $N = 4$, set

$$(g_0, g_1, g_2, g_3, g_4) = (1, 0, 0, 1, 1) = (23)_8.$$

- Maximal period sequences are called ***m*-sequences**

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Commonly Used Feedback Connections

SR Length, N	Feedback Connections (in Octal Format)
2	7
3	13
4	23
5	45, 67, 75
6	103, 147, 155
7	203, 211, 217, 235, 277, 313, 325, 345, 367
8	435, 453, 537, 543, 545, 551, 703, 747

Figure: Non-exhaustive list of feedback connections to obtain m-sequences.

Properties of m-Sequences

- Are periodic with period $= 2^N - 1$
- Contain approximately equal number of ones and zeros in any one period
- **Autocorrelation function** is nearly identical to that of IID $\mathrm{Ber}(0.5)$ process
  - Suppose $X_1, X_2, \cdots \overset{\text{i.i.d.}}{\sim} \mathrm{Ber}(0.5)$. Then,

$$M_X(t) = 0.5, \qquad R_X(s,t) = \mathbb{E}[X_s X_t] = \begin{cases} \frac{1}{2}, & s = t, \\ \frac{1}{4}, & s \neq t. \end{cases}$$

  - Given a discrete-time signal $\{x[n]\}_{n=0}^{\infty}$ with period N, its autocorrelation is given by

$$R_X[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n]\, x[n+k], \qquad k \in \{0, 1, 2, \ldots\}.$$

  - Considering the single period ($N = 15$) output $1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0$, we have

$$R_X[k] = \begin{cases} \frac{8}{15}, & k = 0, \\ \frac{4}{15}, & 1 \leq k \leq 14, \end{cases} \qquad R_X[k+15] = R_X[k].$$

Non-Binary PRNGs

- How do we generate non-binary m-sequences?
- One technique is the **power residue method**
- Given numbers (a, x_0, p), let

$$x_n = a x_{n-1} \bmod p, \qquad n \in \mathbb{N}.$$

  - p is typically a large prime (e.g., $p = 2^{31} - 1 = 2147483647$)
  - $x_0 \neq 0$ is called the **seed** value (e.g., $x_0 = 42$)
  - $a \neq 1$ is the multiplicative factor
- Due to $(\bmod\ p)$ operation, $x_n \in \{1, \ldots, p-1\}$ for all n
- The choice of (a, p) is crucial to obtain an m-sequence

Non-Binary PRNGs

Recursion

$$x_n = a x_{n-1} \bmod p, \qquad n \in \mathbb{N}.$$

- If $(a, p) = (4, 7)$, then output sequence (with $x_0 = 1$) is

$$(1, 4, 2, 1, 4, 2, \cdots)$$

- If $(a, p) = (3, 7)$, then output sequence (with $x_0 = 1$) is

$$(1, 3, 2, 6, 4, 5, 1, 3, 2, 6, 4, 5, \cdots)$$

- In most programming languages:
  - $a = 7^5, \quad p = 2^{31} - 1.$
  - Output normalised to take values in $\left\{\frac{1}{p}, \frac{2}{p}, \ldots, \frac{p-1}{p}\right\}$